[ ALPACA AI TRADING AGENTS · OPTIONS ALPHA AGENTS ]

# THE DESK THAT GRADES ITSELF.

An options desk where **deterministic code builds every trade** and each analyst carries a track record. Analysts state probabilities, not verdicts, are Brier-scored against what actually happened, and one that is confidently wrong **loses its vote**. The models choose between pre-built candidates or refuse; they cannot invent a strike, because no code path allows it.

| 783 | 3,686 | 632 | 1 |
|---|---|---|---|
| TESTS, NO NETWORK | CONTRACTS READ | STRUCTURES BUILT | CHOSEN, OR NONE |

## WHAT ONLY THIS DESK DOES

- **Grades its own analysts.** Brier scores from resolved outcomes set voting weights each cycle. Unproven analysts stay at 1.0; a demoted one floors at 0.2 so it always has a path back.
- **Runs an adversary** whose objections are journalled whether or not they prevail. In a recorded cycle its objection to a thin hedge moved the trader off the highest-credit candidate.
- **Two vetoes that fail differently.** One is pure code checking position delta against the structure's own thesis; the other is a model shown the trade but never the committee's reasoning.
- **Refusal is the product.** Two of four replayable scenarios are refusals.

## ONE REAL DECISION, END TO END

| | |
|---|---|
| snapshot | SPY 769.35, IV 2.27pp over realized |
| vol_analyst | p=0.63 · premium is rich |
| bear_adversary | p=0.40 · breakevens only 0.4% away |
| trader | picks c3, the wider cushion, answering the adversary |
| thesis veto | PASS · net delta -8.46 matches the thesis |
| blind veto | PASS · independent, never saw the debate |
| guard | ALLOW · risk $322, 1 of 3 positions |

The adversary changed the trade. That is the difference between a committee and a rubber stamp.

## IS IT REASONING, OR REMEMBERING?

The standing criticism of LLM trading agents is **knowledge contamination**: papers such as TradingAgents (arXiv 2412.20138) backtest over dates inside the model's own training data, so a good result may be recall.

Our answer is checkable. The model's cutoff is **May 2026**; live decisions run on **August 2026** data. There was nothing to memorise. The walk-forward harness does span earlier dates, and that is acceptable only because it runs no model at all.

## EVIDENCE, AND ITS LIMITS

| SYMBOL | TRADES | WIN | EXPECTANCY | MAX DD |
|---|---|---|---|---|
| SPY | 8 | 87.5% | +0.62R | 2.0R |
| QQQ | 7 | 85.7% | +0.57R | 2.0R |
| AAPL | 8 | 62.5% | -0.12R | 6.0R |
| MSFT | 7 | 85.7% | +0.57R | 2.0R |

> **Read this before the numbers.** Thirty out-of-sample trades proves nothing statistically, and one symbol loses money, which is the point: an earlier version of this harness scaled its risk threshold to the wrong horizon and produced a 97% win rate by construction. It was caught and corrected before publication.

## WHAT WE CANNOT PROVE

- Paper trading only. No slippage or fill-quality evidence.
- Few closed live trades, so most analyst weights are still 1.0. Demoting on a handful of outcomes is the error we refuse to make.
- The fail-closed scenario is constructed, and its fixture says so.

## VERIFY IT YOURSELF

- `scripts/replay.py --all --verify` reproduces four recorded verdicts offline, with no credentials.
- `make verify-journal` checks the hash chain.
- Orders route through the official Alpaca CLI; market data through the official MCP server with **no trading toolset exposed**.